EE3032 Introduction to VLSI Design—Homework 1

1. **MOS Theory** (2 points)
   Fig. 1 shows a CMOS inverter. Assume that the body effect can be neglected. Identify the transistor(s) which cannot operate in the saturation region. Explain your answer.

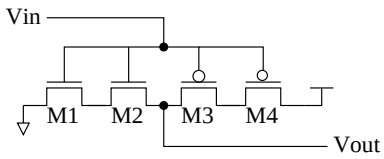

Figure 1: A CMOS inverter

2. **MOS Theory** (2 points)
   Show that the series connection of MOS transistors shown in Fig. 2 behaves as a single MOS transistor with twice the length of the individual MOS transistors. Neglect the body effect.

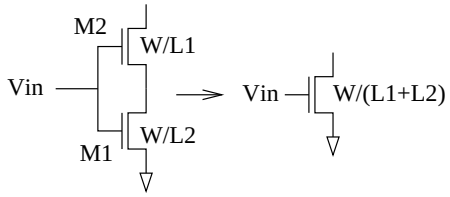

Figure 2: MOS transistors operating in series

3. **CMOS Design** (3 points)
   According to *Shannon's expansion theorem*, a Boolean function with n variables can be expanded on any one of variable x_i. That is, $f(x_1, x_2, \ldots, x_i, \ldots, x_n) = x_i \cdot f(x_1, x_2, \ldots, 1, \ldots, x_n) + \bar{x_i} \cdot f(x_1, x_2, \ldots, 0, \ldots, x_n)$. This implies that we can realize a Boolean function using multiplexers. Fig. 3 shows an example of a two-input AND gate realized by a 2-to-1 multiplexer. Use a 4-to-1 multiplexer to implement $F = \bar{A}(B + \bar{C}) + A\bar{B}C$.

A	B	F
0	0	0
0	1	0
1	0	0
1	1	1

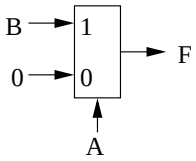

Figure 3: Using multiplexer to realize a two-input AND gate

4. **Static CMOS Logic** (4 points)
   Realize each of the following Boolean functions using static CMOS logic: (a) $f(w, x, y, z) = \overline{w \cdot \overline{(x + y)} + y \cdot z}$ (b) $f(x, y, z) = \bar{x}\bar{y} + \bar{x}\bar{z} + \bar{y}\bar{z}$. Suppose that only true literals are available. You should show the transistor diagram.